

JAMES DE RAJA

Senior Real-Time Performance Engineer · Unity Rendering · XR · Online Multiplayer
Chennai, India · Open to remote worldwide & relocation · jamesderaja@gmail.com · [Portfolio](#) · [LinkedIn](#) · [GitHub](#)

Senior Unity Performance Engineer with 13+ years optimizing real-time games and large-scale systems. Improved performance **4× at 10K entities** and optimized systems handling **5M+ executions/day**. Architecting an open-world shooter with 10K+ entities, multiplayer systems, and XR rendering pipeline. See detailed case studies at james.alphaden.club

Core Strengths:

- Real-time Performance Optimization
- Large-Scale Simulation
- XR Rendering

TOP RESULTS

- **40 ms → 9.4 ms** · 4× frame-time reduction at 10K entities under heavy load (ECS architecture, ~24 → ~106 FPS, no visual change). [View Case Study](#)
- Identified critical GPU bottleneck in XR rendering (+7.27 ms overdraw cost — 201 transparent draws, stereo 90 Hz). [View Case Study](#)
- **66 → 136 FPS** · 10K-mesh GPU instancing; CPU render-thread time collapsed, render no longer CPU-bound. [View Lab](#)

CURRENT FOCUS — OPEN-WORLD SHOOTER

Update Strategies at Scale — ECS Architecture · [View Case Study](#) · [View Repo](#)

- **4× frame-time reduction (40 → 9.4 ms; 24 → 106 FPS)** at high-density enemy scenes — no visual change.
- Evaluated OOP, manager, and ECS architectures to identify optimal scaling strategy for large-scale simulation.

XR Stress Lab — Unity 6 / URP / OpenXR · [View Case Study](#) · [View Repo](#)

- Built internal tooling to identify GPU bottlenecks in XR rendering (overdraw, MSAA, rendering cost).
- **+7.27 ms GPU delta** from 201 transparent draws (stereo 90 Hz); GPU cost doubled when overdraw compounded with MSAA 4×. Findings now define the shooter's XR rendering strategy.

GPU Instancing Lab · [View Lab](#)

- **66 → 136 FPS (15.04 → 7.34 ms)** at 10K identical meshes; CPU render-thread collapsed, now default for enemy archetypes.

PRODUCTION GAME WORK

Sneaky Warrior 3D — Frame-Budget Stabilization · [View Case Study](#) · [App Store](#)

- **Held 16 ms across mobile tiers including tile-based GPUs** with 100+ animated enemies, ragdolls, and projectile bursts.
- Optimized large-scale enemy rendering using batching and spatial grouping while preserving culling efficiency.
- Off-screen skinning gated; performance spikes profiled and mitigated. Patterns now applied in the open-world shooter.

8-Player Real-Time Multiplayer — 25K+ Downloads

- **Built real-time multiplayer system** delivering stable 60 FPS across 8 players under input-saturation stress.
- Designed reusable multiplayer architecture now used in the current large-scale project.

Supersonic Hypercasual Competition — Round 1 Winner (\$2M prize pool)

- Selected from a global pool judged on CPI, D1 retention, and FTUE; shipped through Supersonic's live-ops pipeline.
- Repeat publisher relationships with Voodoo, Lion Studios, and Supersonic across 50+ commissioned prototypes and 100+ public releases. [View Portfolio](#)

DISTRIBUTED SYSTEMS ENGINEERING

Member Leadership Staff (Staff Engineer) · Zoho Corporation Jun 2017 – Present · Chennai

9 years building high-scale distributed systems with strict uptime and reliability requirements.

- **Led performance-focused system redesigns** enabling 99.9% uptime and zero-downtime migrations across high-scale infrastructure (5M+ daily executions).
- **Owned and scaled Zoho CRM integration** powering 5M+ daily executions across 8,000+ organizations. Leading large-scale system migration with zero customer disruption.
- Built 5 of the top 10 most-used integrations on Zoho Flow; 100+ integrations shipped overall.
- Architected a **zero-downtime environment-variable system for Zoho Creator** — 1M+ executions/day across 1,500+ orgs, zero customer-reported incidents post-release.
- **Led remediation of a production server-crash incident:** reduced system load 70–85% with API contract unchanged (3M executions/day, 2,500+ orgs).
- Mentor 6+ active mentees, including 2 direct reports; total mentorship footprint of 20+ engineers across Zoho product teams.

ENGINEERING PRACTICE

Approach: profiler-driven optimization, CPU/GPU bottleneck isolation, and data-oriented design (ECS/DOTS).

TECHNICAL SKILLS

Rendering & Performance: Unity (URP, Built-in), ECS / DOTS, GPU Instancing, Frame Debugger, GPU Timing Analysis, RenderDoc

XR & Systems: OpenXR, Unity XR Toolkit, Stereo Rendering Optimization, and real-time performance optimization

Profiling: Unity Profiler, RenderDoc, GPU Timing Queries, custom runtime overlay

Backend: Java, Distributed REST APIs / Webhooks, High-scale systems (5M+ executions/day), Git, GitHub Actions

EDUCATION

B.E., Electronics & Communication — SSN College of Engineering · 2013-2017